Ecological Network Structure Is Multidimensional: Rethinking the Stability Debate

Tanya Strydom [1]; Andrew P. Beckerman [1]

Abstract: Ecological networks are commonly characterised using a wide array of structural metrics, yet there remains little consensus on how these descriptors relate to ecological stability. Different studies frequently identify different structural predictors of stability, leading to long-standing debate about the role of network architecture in ecosystem dynamics. One possible explanation is that network structure is not a single property but a multidimensional feature of ecological systems. If so, commonly used metrics may capture different aspects of network organisation rather than alternative measures of the same underlying attribute. Here we test this idea by examining the relationships among thirty widely used food-web metrics across a compilation of empirical networks. We show that these descriptors organise into statistically robust modules that correspond to distinct dimensions of network architecture. Multivariate analyses further reveal that food webs occupy a low-dimensional structural space despite the large number of available metrics. Many commonly used descriptors capture overlapping structural information, whereas others represent independent axes of variation in how trophic interactions are organised. These results suggest that apparent inconsistencies in the structure–stability literature may arise because different studies measure different dimensions of network structure. Rather than asking which single metric predicts stability, analyses of ecological networks may benefit from explicitly considering the multidimensional nature of network architecture. By identifying the major structural domains underlying food-web organisation, this study provides a conceptual framework for interpreting network metrics and for more systematically linking network structure to ecological dynamics.

Keywords: food web, structure, dimensionality reduction

1 Introduction

Which metrics best represent the distinct stages of the energy-flow hierarchy, and do they capture different components of stability?

Food web structure self-organises into process-level modules, and those modules govern different components of stability.

Food web architecture is organised into emergent structural process modules, and different modules regulate different components of ecological stability.

Food web metrics self-organise into structural process modules, these modules define network space, and different modules regulate different stability components.

Should we be looking at if different metrics scale predictably with each other?

Emergent property (thinking like the cool tensor work/dynamic models) vs static metrics (using one measure of structure)

Ecological networks are commonly characterised using a large and diverse set of structural metrics, yet there remains little consensus on how these metrics should be interpreted or compared across studies (Lau et al., 2017; Vermaat et al., 2009). Measures of connectance, trophic level, modularity, omnivory, motif structure, and many others are routinely invoked as indicators of ecosystem organisation or stability. However, these descriptors often covary strongly, capture overlapping information, or reflect different scales of network organisation. As a result, studies using different subsets of metrics frequently arrive at conflicting conclusions about the relationship between network structure and ecological stability.

One explanation for this inconsistency is that network structure is fundamentally multidimensional. Rather than representing a single property of ecosystems, structural metrics may capture different aspects of how interactions are organised and how energy moves through ecological systems. If this is the case, then individual metrics should not be expected to provide a universal predictor of stability. Instead, they may represent distinct dimensions of structural organisation that relate to different ecological mechanisms. Despite this possibility, most studies treat structural descriptors independently or select a small subset based on convention or data availability. Relatively few attempts have been made to examine whether commonly used network metrics organise into a smaller set of coherent structural dimensions that define the underlying space of ecological network architecture.

Ecological stability is likewise not a single property but a collection of related dynamical behaviours [Pimm 1984; Ives & Carpenter (2007)]. Different studies have emphasised different stability components, including the

persistence of species [REF], the resistance of communities to disturbance [REF], and the speed with which systems recover following perturbation [REF]. These processes reflect distinct aspects of ecosystem dynamics and may depend on different features of network structure. From an energy-flow perspective, three stability mechanisms are particularly relevant: persistence, resistance, and return. Persistence refers to the continued maintenance of energy flow and species presence following species loss or environmental change. Resistance describes the degree to which disturbances propagate through ecological interactions, potentially amplifying or dampening perturbations. Return captures the capacity of a system to reorganise and re-establish stable interaction patterns following disturbance. Because these mechanisms operate through different ecological processes, they may be expected to depend on different aspects of network architecture. Structural features related to species roles and redundancy may influence persistence, whereas properties governing interaction pathways and network organisation may affect disturbance propagation or recovery dynamics.

Within this framework, metrics that are often treated as competing predictors of 'stability' instead emerge as complementary descriptors of different stability mechanisms (Thompson et al., 2012). Node-level metrics primarily relate to persistence, path-level metrics to resistance, and global organisational metrics to return dynamics. Some descriptors span multiple scales, reflecting the coupling between structural organisation and emergent behaviour (Allesina & Tang, 2012). This perspective provides a mechanistic explanation for why studies using different network metrics frequently report contrasting structure–stability relationships. Rather than reflecting inconsistency or redundancy, these differences arise because different metrics implicitly target different components of stability (Lau et al., 2017). By explicitly linking structural scale, energy flow, and stability mechanism, this framework provides a principled basis for interpreting network metrics and for selecting descriptors that align with specific ecological questions.

Here we investigate whether the large number of structural metrics used to characterise food webs can be reduced to a smaller set of coherent structural dimensions, and whether these dimensions relate to different components of ecological stability. Using a compilation of empirical food webs, we quantified a broad suite of commonly used network descriptors spanning species roles, interaction pathways, and global network organisation. We first examined whether these metrics self-organise into statistically robust modules based on their covariation across ecosystems. We then evaluated how these modules align with dominant multivariate axes of structural variation. Finally, we compared alternative representations of network structure to assess how different structural dimensions relate to three stability properties: robustness to species loss, spectral radius, and structural complexity. By explicitly mapping the multivariate structure of network architecture to stability outcomes, this analysis provides a principled framework for interpreting network metrics and for selecting structural descriptors that match specific ecological questions.

2 Materials & Methods

2.1 Data Compilation

We compiled quantitative network data from XX, resulting in a total of XX ecological networks. Each network was characterized using a suite of XX structural metrics Table 1, including classic descriptors such as richness, connectance, and modularity, as well as information-theoretic measures like spectral radius and SVD complexity. Prior to analysis, networks with missing values were omitted, and all metrics were standardized (mean = 0, SD = 1) to account for differences in scale and units across descriptors.

Table 1: An informative caption about the different network properties. We use a combination of metrics from both the original Vermaat et al. (2009) paper as well as including those that have been identified by Thompson et al. (2012) and have been linked to emerging ecosystem properties such as stability

Label	Definition	Structural interpretation	Reference
Basal	Proportion of taxa with zero vulnerability (no consumers).	Quantifies the proportion of species representing basal energy inputs to the network.	
Top	Proportion of taxa with zero generality (no resources).	Describes the relative prevalence of terminal consumers in the network.	
Intermediate	Proportion of taxa with both consumers and resources.	Captures the proportion of species participating in both upward and downward energy transfer.	
Richness (S)	Number of taxa (nodes) in the network.	Describes network size.	
Links (L)	Total number of trophic interactions (edges).	Describes interaction density independent of network size.	
Connectance	L/S^2, where S is the number of species and L the number of links	Measures the proportion of realised interactions relative to all possible interactions.	Dunne et al. 2002

Label	Definition	Structural interpretation	Reference
L/S	Mean number of links per species.	Captures average interaction density per taxon.	
Cannibal	Proportion of taxa with self-loops.	Quantifies the prevalence of cannibalistic interactions.	
Herbivore	Proportion of taxa feeding exclusively on basal species.	Describes the representation of primary consumers.	
Intermediate	Percentage of intermediate taxa (with both consumers and resources)		
Trophic level (TL)	Prey-weighted trophic level averaged across taxa.	Captures the vertical organisation of energy transfer.	Williams & Martinez (2004)
MaxSim	Mean maximum trophic similarity of each taxon to all others.	Quantifies functional similarity based on shared predators and prey.	Yodzis & Winemiller (1999)
Centrality	Node centrality averaged across taxa (definition-dependent).	Captures the distribution of influence or connectivity among species.	Estrada & Bodin (2008)
ChLen	Mean length of all food chains from basal to top taxa.	Describes the average number of steps in energy-transfer pathways.	
ChSD	Standard deviation of food chain length.	Captures variability in pathway lengths.	
ChNum	Log-transformed number of distinct food chains.	Quantifies the multiplicity of alternative energy pathways.	

Label	Definition	Structural interpretation	Reference
Path	Mean shortest path length between all species pairs.	Describes the average distance between taxa within the network.	
Diameter	Maximum shortest path length between any two taxa.	Captures the largest network distance between species.	
Omnivory	Proportion of taxa feeding on resources at multiple trophic levels.	Describes vertical coupling of energy channels.	McCann (2000)
Loop	Proportion of taxa involved in trophic loops.	Quantifies the prevalence of cyclic interaction pathways.	
Prey:Predator	Ratio of prey taxa (basal + intermediate) to predator taxa (intermediate + top).	Describes the overall shape of the trophic structure.	
Diameter	Diameter can also be measured as the average of the distances between each pair of nodes in the network		
Clust	Mean clustering coefficient.	Measures the tendency for taxa sharing interaction partners to also interact with each other.	Watts & Strogatz (1998)
GenSD	Normalised standard deviation of generality.	Captures heterogeneity in the number of resources per taxon.	Williams & Martinez (2004)
VulSD	Normalised standard deviation of vulnerability.	Captures heterogeneity in the number of consumers per taxon.	Williams & Martinez (2004)

Label	Definition	Structural interpretation	Reference
LinkSD	Normalised standard deviation of total links per taxon.	Quantifies variation in species connectivity.	
Intervality	Degree to which taxa can be ordered along a single niche dimension.	Measures the extent of niche ordering in trophic interactions.	Stouffer et al. (2006)
(Spectral radius)	Largest real part of the eigenvalues of the undirected adjacency matrix.	Captures a global property of network organisation related to interaction strength aggregation.	
Complexity (SVD)	Shannon entropy of the singular value decomposition of the adjacency matrix.	Quantifies heterogeneity in interaction structure.	Strydom et al. (2021)
Robustness	Proportion of secondary extinctions following primary species removal.	Operational measure of tolerance to node loss.	Jonsson et al. (2015)
S1 (Linear chain)	Frequency of three-node linear chains (A $\rightarrow$ B $\rightarrow$ C) with no additional links.	Captures the prevalence of simple, unbranched energy-transfer pathways.	Stouffer et al. (2007) Milo et al. (2002)
S2 (Omnivory)	Frequency of three-node motifs forming a feed-forward loop (A $\rightarrow$ B $\rightarrow$ C, A $\rightarrow$ C).	Describes vertical coupling of trophic levels within small subnetworks.	Stouffer et al. (2007) Milo et al. (2002)
S4 (Apparent competition)	Frequency of motifs where one consumer feeds on two resources (A $\rightarrow$ B $\leftarrow$ C).	Captures the prevalence of shared-predator structures among resources.	Stouffer et al. (2007) Milo et al. (2002)

Label	Definition	Structural interpretation	Reference
S5 (Direct competition)	Frequency of motifs where two consumers share a single resource (A $\leftarrow$ B $\rightarrow$ C).	Describes the occurrence of shared-resource structures among consumers.	Stouffer et al. (2007) Milo et al. (2002)

2.2 Identification of Structural Modules

We tested the hypothesis that structural descriptors of food webs are organised into statistically coherent modules reflecting shared ecological function or scaling relationships, rather than forming arbitrary clusters driven by sampling noise. If such modular organization exists, then metrics within a module should exhibit strong internal correlation relative to metrics assigned to different modules. The resulting clusters should be robust to resampling of the data. The identified modular structure should exceed expectations under null models of random association among metrics.

We quantified pairwise associations among the 30 structural metrics using Pearson correlations computed across food webs. Because ecological metrics may covary either positively or negatively depending on scaling relationships, we constructed two alternative distance matrices: $1 - r$, which preserves the sign of correlations and distinguishes positive from negative association and $1- \mid r \mid$, which groups variables based on the magnitude of their association regardless of sign. These two definitions allow us to test whether modular structure depends on directional relationships or simply on the strength of coupling among metrics. Hierarchical clustering was performed using average linkage on each distance matrix to identify candidate structural modules.

The optimal number of clusters was evaluated across a range of partition sizes (k = 2–10) using average silhouette width to assess within-cluster cohesion and between-cluster separation. To further evaluate cluster robustness, we implemented bootstrap resampling with 1,000 bootstrap replicates. This procedure estimates approximately unbiased (AU) p-values for each cluster, quantifying the probability that a cluster is supported under repeated resampling of the data. Clusters were considered statistically robust when they exhibited high silhouette support, and approximately unbiased bootstrap support 0.95. This dual criterion ensures that identified modules are both structurally coherent and stable to sampling variation.

2.3 Multivariate Structure and Module–Axis Alignment

To evaluate whether structural metrics of food webs organize into coherent multivariate modules and whether those modules define principal axes of variation, we conducted a principal component analysis (PCA) followed by a permutation-based test of module–axis alignment. Under this framework, principal components define dominant structural gradients in the metric space, whereas modules represent hypothesised mechanistic groupings of structurally related metrics. Demonstrating alignment between these two representations would suggest that modular decomposition captures fundamental axes of ecological variation.

Skewed count-based metrics (*e.g.,* link counts, interval counts, and related quantities) were log-transformed using $log(x + 1)$ to reduce right skew. All remaining variables were standardized to zero mean and unit variance prior to analysis to ensure that metrics measured on different scales contributed equally to the ordination.

To quantify how structural modules align with principal axes, we decomposed variance in each principal component according to module membership. For each module m and principal component k, we computed:

$$A_{mk} = \sum_{i \in m} l_{ik}^2$$

where l_{ik} is the loading of metric i on principal component k and A_{mk} represents the fraction of variance in PC k attributable to metrics in module m. Because squared loadings sum to one within each principal component, this provides a direct partitioning of PC variance across modules. This produces a module $\times$ PC matrix describing geometric alignment between modular structure and multivariate axes.

To evaluate whether observed module–PC alignment exceeded expectations under random module structure, we implemented a permutation test. We randomly permuted module labels among metrics 1 000 times while preserving the number and size distribution of modules, and the PCA loadings. For each permutation r, we recomputed $A_{mk}^{(r)}$ so as to form a null distribution for each module–PC pair. From this we computed p-values and corresponding z-scores, which was then used to infer significant alignment when $p_{mk} < 0.05 \wedge |Z_{mk}| > 1.96$

$$p_{mk} = P(A_{mk}^{(r)} > A_{mk})$$

$$Z_{mk} = \frac{A_{mk} - \mu_{mk}}{\sigma_{mk}}$$

Additionally we also evaluated if modules collectively aligned with PCA structure beyond random expectation. Overall concentration of variance within module–PC space was calculated as $T = \sum_{m,k} A_{mk}^2$ and significance was determined by comparing the observed (T) the permutation distribution ($T^{(r)}$). Where $p_{global} = P(T^{(r)} \geq T)$. A significant result indicates that module structure explains multivariate variance better than random groupings.

2.4 Structural Metric Reduction and Representation of Network Space

To evaluate how network structure predicts ecological stability, we first reduced a high-dimensional set of topological metrics into alternative, biologically interpretable predictor sets. Because dimensionality reduction can fundamentally shape inference, we explicitly compared two conceptually distinct representations of network structure: (1) structural domain representatives and (2) latent axes of network variation.

2.4.1 Identification of Structural Domains

We previously grouped network metrics into clusters based on their pairwise correlations, such that each cluster represented a structurally coherent domain (*e.g.,* trophic composition, centralization, path structure). Clustering was performed using correlation-based distance, ensuring that metrics grouped together reflected shared structural information rather than raw scale. To construct a reduced predictor set from these domains, we selected a single representative metric per cluster using a medoid approach. Within each cluster, we computed the absolute correlation matrix among member metrics and defined distance as 1− r . The medoid was identified as the metric minimizing mean pairwise distance to other metrics in the cluster. This approach preserves structural diversity while minimizing redundancy and does not rely on principal component loadings, which can bias selection toward dominant variance axes. The resulting 'cluster medoid' set retained one interpretable metric per structural domain. Importantly, this procedure prioritizes preservation of structural domains rather than overall variance magnitude, allowing low-variance but potentially mechanistically relevant features of network topology to be retained.

2.4.2 Extraction of Latent Structural Axes

As a complementary representation of network structure, we performed principal component analysis (PCA) on the scaled full metric matrix. Principal components were retained until cumulative explained variance exceeded 80%, yielding a set of orthogonal axes describing the dominant gradients of variation in network topology. These retained PC scores constitute a low-dimensional latent representation of network space. Unlike cluster medoids, which preserve structural categories, PC scores preserve maximal variance and ensure orthogonality among predictors. We also explored a third reduction approach—selecting the single highest-loading metric

per retained PC. However, because principal components are linear combinations of multiple metrics, this "PC-dominant metric" strategy substantially reduces the variance represented by each axis. Consequently, this approach captures less of the total structural variance and was treated as exploratory rather than primary.

2.4.3 Conceptual Contrast Between Representations

The cluster-medoid and PCA-score representations reflect different theoretical assumptions about how structure relates to stability. The cluster-medoid approach assumes that ecological stability responds to discrete structural domains that may vary independently and need not align with the dominant gradients of network variation. In contrast, the PCA-score approach assumes that stability responds primarily to the major axes of variation in network topology, regardless of their interpretability in terms of structural domains. By explicitly comparing predictive performance across these representations, we test whether ecological stability is better explained by domain-specific structural mechanisms or by global geometric gradients of network organization.

2.4.4 Structural Control of Stability

To evaluate how different hierarchical representations of network structure explain variation in stability, we implemented a regularized linear modelling framework designed to balance interpretability and overfitting risk given the modest sample size (n = 38). For each stability component (complexity, robustness, and), we fit elastic net regression models using three distinct structural representations: Fine-scale cluster-level topological descriptors, Dominant structural summaries (PC_Dominant), and Multivariate principal component scores (PC_Scores). All predictors and response variables were standardized prior to analysis to allow direct comparison of coefficient magnitudes.

Elastic net regularization was chosen because it permits inference under correlated predictors by interpolating between ridge (distributed shrinkage) and lasso (sparse selection) penalties. The mixing parameter (0–1) determines the degree of sparsity, with lower values indicating distributed structural influence and higher values indicating concentration onto a smaller subset of predictors.

2.4.4.1 Model Evaluation

Predictive performance was assessed using repeated v-fold cross-validation (5 folds × 20 repeats). Within each training partition, models were fit across a grid of values (0–1 in increments of 0.25), and the optimal penalty strength ( ) was selected via internal cross-validation. Performance was quantified as cross-validated R^2 computed on held-out test partitions. After cross-validation, final models were refit to the full dataset using

the mean selected , and standardized coefficients were extracted to characterize the direction and magnitude of structural effects. This approach was not intended to maximize predictive accuracy per se, but rather to quantify the relative degree and architecture of structural control across stability facets.

3 Results

3.1 Structural Metrics Organise into Robust Modules

Hierarchical clustering of structural descriptors revealed clear modular organisation within food-web architecture Figure 1. Silhouette analysis indicated an optimal partition of the signed correlation matrix at k=7 modules, with a maximum average silhouette width of [insert value]. Clustering based on absolute correlations produced a slightly coarser but comparable solution (k=5), demonstrating that the modular structure is robust to the treatment of correlation sign and reflects strong underlying association patterns among metrics.

[Figure 1 about here.]

Bootstrap resampling further supported the stability of the major clusters. Most principal modules exhibited high approximately unbiased (AU) support values (AU [insert value]), indicating that the identified clusters are not artefacts of sampling variability but represent statistically robust groupings of structural descriptors.

The resulting modular partition identified coherent groups of metrics describing distinct aspects of network architecture. These included a large-scale architectural module integrating size and motif-based descriptors, modules capturing trophic organisation and energy routing, and two modules consisting of isolated structural descriptors that function independently from broader metric groupings. Importantly, the modular structure persisted after accounting for species richness, demonstrating that clustering was not driven solely by network size effects.

3.2 Modular Decomposition of Structural Space

The seven-module solution revealed a hierarchical decomposition of structural space into interpretable architectural dimensions. Collectively, the module decomposition reveals that food web structure can be interpreted as statistically supported and ecologically interpretable dimensions of network organisation that is strongly suggestive of a delineation by hierarchical architecture.

Macro-Architectural Complexity: The largest module encompassed species richness, number of interactions, diameter, motif frequencies, intervality, and related descriptors. These metrics jointly describe large-scale expansion of the interaction network and increased combinatorial complexity as species richness

increases. This module reflects macro-architectural scaling: increasing network size simultaneously expands motif diversity, path structure, and global trophic ordering.

Trophic Integration: A second module grouped metrics describing vertical trophic organisation and connectivity, including connectance, trophic level, omnivory, chain length, and clustering. Together, these descriptors quantify how densely and hierarchically energy and interactions are embedded across trophic strata. This module captures the degree to which the network integrates trophic levels through cross-level interactions and structural embedding.

Energy Transport: A distinct module isolated geometric descriptors of interaction pathways, including mean distance, path number, and dispersion in chain length. These metrics characterize the structural geometry of energy routing independent of overall network size. The separation of this module suggests that routing structure constitutes an independent dimension of architectural variation rather than merely scaling with complexity.

Trophic Asymmetry: Another module captured metrics describing structural imbalance between basal and consumer components, including the proportion of basal species, predator–prey ratios, and generality measures. This module reflects bottom-heavy versus top-heavy structural skew and quantifies asymmetry in the distribution of trophic roles across the network.

Control Heterogeneity: A separate module grouped descriptors of heterogeneity in consumer pressure and top-down control, including the proportion of apex species, herbivory structure, and variance in vulnerability. This module captures variation in how evenly or unevenly control is distributed across trophic levels.

Centralisation and Functional Redundancy: Two metrics formed independent modules. Centrality emerged as a standalone module, representing concentration of structural dominance within a subset of species. Maximum trophic similarity likewise formed an isolated module, quantifying functional redundancy and niche overlap. The separation of these descriptors from broader structural groupings indicates that dominance concentration and trophic redundancy represent independent axes of food-web organisation.

Collectively, these seven modules delineate a hierarchical decomposition of food-web structure spanning macro-scale complexity, vertical integration, routing geometry, trophic imbalance, control heterogeneity, dominance concentration, and functional overlap.

Table 2: Structural Modules Identified From Hierarchical Clustering.

Module Name	Metrics Included	Ecological Interpretation
Macro-Architectural Complexity	richness, links, diameter, motif frequencies, intervality, path-related metrics	Captures large-scale expansion of network structure and combinatorial complexity associated with increasing system size
Trophic Integration	connectance, trophic level, omnivory, chain length, clustering	Describes vertical trophic embedding and density of cross-level interactions
Energy Transport	mean distance, path number, chain-length dispersion	Quantifies geometric structure of energy routing independent of size scaling
Trophic Asymmetry	basal proportion, predator–prey ratio, generality	Represents structural skew between basal and consumer dominance; bottom-heavy vs top-heavy organization
Control Heterogeneity	apex proportion, herbivory, vulnerability variance	Captures heterogeneity in top-down control and uneven distribution of interaction pressure
Centralisation	centrality	Measures concentration of structural dominance within a subset of species
Functional Redundancy	maximum trophic similarity	Quantifies trophic overlap and niche redundancy; independent axis of functional equivalence

3.3 Structural Modules Align with Dominant Axes of Network Variation

Principal component analysis of standardised structural metrics revealed strong dimensional compression in food-web structure Figure 2. The first three principal components accounted for 69.1% of total variance (PC1 = 31.2%, PC2 = 22.4%, PC3 = 15.4%), with 80.8% of variance explained by the first five components. Subsequent axes contributed progressively smaller proportions of variance, indicating that a small number of dominant gradients summarise most structural variation.

[Figure 2 about here.]

[Figure 3 about here.]

Inspection of the loading structure showed that the first principal component primarily reflected variation in network size and connectivity. Metrics such as richness, number of links, diameter, path length, and measures associated with topological complexity loaded strongly on this axis. The second principal component captured variation associated with structural organization and connectivity patterns, whereas the third component reflected variation in trophic structure and vertical differentiation, with strong contributions from trophic level, chain length, and related measures. Together, these axes represent interpretable structural gradients in the metric space.

[Figure 4 about here.]

The global permutation test indicated significant alignment between the modular partition and the principal component structure ($P = 0.005$), demonstrating that the observed module assignment explains the multivariate geometry of structural metrics better than randomly assigned modules. Alignment strength varied substantially across modules Figure 5. Macro Complexity (Module 1) exhibited the strongest and most extensive alignment, showing significant associations with five principal components and a dominant loading on PC1, indicating that it contributes substantially to the primary structural gradient associated with network size and connectivity. Energy Transport (Module 3) showed significant alignment with four principal components, with its strongest association on PC4, suggesting that it captures structural variation largely orthogonal to the dominant complexity axis. Trophic Integration (Module 2), Trophic Asymmetry (Module 4), Control Heterogeneity (Module 5), and Functional Redundancy (Module 7) displayed more limited alignment, each significantly associated with one to two principal components, indicating more constrained mapping onto the structural feature space. Notably, Functional Redundancy exhibited its strongest association with a high-order principal component (PC7), suggesting sensitivity to more subtle structural variation. Centralisation (Module 6) showed no significant alignment under the joint permutation and Z-score criterion, indicating that it does not meaningfully structure variation across the retained principal components. Overall, significant module–PC associations were concentrated in a subset of principal components, reinforcing the interpretation that modular structure maps onto specific structural gradients rather than uniformly spanning the full multivariate space.

[Figure 5 about here.]

Collectively, these results demonstrate that food-web structural metrics exhibit strong low-dimensional organisation and that independently identified structural modules are non-randomly aligned with the dominant axes of variation. The significant global alignment supports the interpretation that modular decomposition

reflects meaningful geometric structure rather than clustering artifacts. Moreover, the dominance of a single module across multiple principal components suggests hierarchical organisation, wherein a core structural module governs the primary gradient of variation while additional modules capture secondary and specialised structural dimensions.

3.4 Predictors of stability

3.4.1 Dimensional Reduction Characterisation

Representation	Dimensionality	Variance Preserved	Structure Type
Medoids	7	23%	Domain sampling
PC-dominant metrics	4	13%	Axis proxy sampling
PC scores	5	80%	True latent space

To evaluate how alternative structural representations differed in their information content and redundancy, we quantified variance retention, internal correlation structure, effective dimensionality, and geometric similarity among representations.

Variance Retention: The three representations differed substantially in the proportion of total structural variance preserved (Fig. S1A). The cluster-medoid representation retained ~23% of the total variance in the full metric space despite reducing dimensionality to seven predictors. In contrast, the PC-dominant metric set retained only ~13% of total variance, reflecting the fact that individual metrics capture only a fraction of each principal component's multivariate structure. As expected by construction, the retained PC-score representation preserved approximately 80% of total variance. Thus, domain-based reduction preserved more distributed structural information than selecting one metric per PC axis, whereas the PC-score representation maximally preserved dominant gradients of network variation.

Internal Redundancy: Representations also differed in their internal correlation structure (Fig. S1B). PC scores were orthogonal by definition (mean |r| 0), indicating complete statistical independence among predictors. In contrast, cluster medoids exhibited moderate residual correlation (mean |r| = X), suggesting partial overlap among structural domains. PC-dominant metrics showed comparable (or higher/lower — insert result) redundancy relative to medoids. These differences indicate that the three approaches vary not only in information retention but also in predictor independence.

Effective Dimensionality: We next quantified effective dimensionality as the number of axes required to explain 80% of variance within each reduced predictor set (Fig. S1C). The PC-score representation required

X axes, reflecting its design to capture dominant structural gradients. The cluster-medoid representation required X axes to reach the same threshold, indicating that despite containing seven predictors, structural variation was concentrated along fewer effective dimensions. The PC-dominant set exhibited the lowest effective dimensionality (X axes), consistent with its reduced variance retention. Together, these results show that dimensional compression differed across approaches not only in magnitude but in the distribution of variance across axes.

Overall, the three structural representations differed substantially in variance retention, redundancy, and effective dimensionality. Cluster medoids preserved moderate variance while maintaining domain interpretability, PC-dominant metrics retained minimal total variance, and PC scores preserved dominant structural gradients while ensuring predictor orthogonality. These differences establish that the representations encode distinct aspects of network topology, justifying their empirical comparison in predictive analyses of stability.

[Figure 6 about here.]

3.4.2 Predictive Performance Across Structural Representations

3.4.2.1 Predictor Importance and Structural Drivers

Across all stability components, elastic net regularization revealed that structural predictors contributed differently depending on how network space was represented. The degree of regularization ( ) and the distribution of non-zero coefficients varied systematically across representations, indicating that structural abstraction alters which aspects of topology emerge as dominant drivers. Models based on fine-scale cluster descriptors retained multiple predictors with moderate standardized effects, reflecting distributed structural control across correlated topological features. In these models, structural heterogeneity metrics, similarity measures, and connectivity descriptors jointly contributed to explaining variation in stability components. By contrast, models using dominant structural metrics (PC_Dominant) tended to concentrate predictive weight onto a smaller subset of variables. This suggests that dimensional compression isolates key structural features that summarize broader topological variation, albeit at the cost of omitting more nuanced structural contributions. When stability was modelled using principal component scores (PC_Scores), structural influence was reorganized into orthogonal latent axes. In this representation, predictors frequently collapsed onto one or two dominant components, indicating that stability responses align with coordinated structural gradients rather than individual metrics. Together, these patterns demonstrate that the identification of "important" structural drivers is contingent on how network space is represented. Predictor importance is therefore not an intrinsic property of a metric alone, but emerges from the structural abstraction applied.

[Figure 7 about here.]

3.4.2.2 Structural Drivers of Stability

Predictive performance differed substantially across stability components and structural representations, revealing variation in how strongly each facet of stability is encoded in static network architecture. Across all representations, exhibited the strongest structural determination, with the highest cross-validated predictive performance under the multivariate PC representation. This indicates that is most effectively captured as a response to coordinated structural gradients embedded within the network topology. Complexity showed intermediate structural predictability, with highest performance at the fine-scale cluster level. This suggests that complexity is shaped by distributed topological features that are partially lost under structural compression but captured when network heterogeneity is described at higher resolution. Robustness exhibited comparatively weak structural predictability across representations. Although modest improvements were observed under principal component abstraction, overall explanatory power remained low relative to other stability components. This pattern implies that robustness is not strongly determined by static structural descriptors alone and may depend on additional dynamical or context-dependent processes not captured in topology. Importantly, shifts in predictive performance across representations indicate that structural abstraction alters the apparent strength of structural control. Certain stability components become more predictable when topology is expressed as fine-grained metrics, whereas others align more strongly with latent structural gradients. Thus, the representation of network space directly shapes inferences about structural drivers of stability.

3.4.3 Differential Effects Across Stability Components

The relationship between network structure and stability was not uniform across components but instead varied systematically in both magnitude and architecture. Complexity was governed by distributed structural influence, with intermediate regularization and multiple retained predictors across representations. This indicates that complexity reflects the cumulative effect of multiple interacting topological features rather than a single dominant driver. In contrast, displayed stronger alignment with multivariate structural axes. Its predictive structure was most coherent under principal component abstraction, suggesting that it emerges from integrated patterns of connectivity and organization that span multiple network dimensions. Robustness differed qualitatively. Under principal component representation, models shifted toward sparsity, with a single dominant structural axis explaining most retained signal. However, overall explanatory power remained modest. This implies that robustness may be only weakly constrained by static network architecture and potentially more sensitive to perturbation dynamics or species-level functional traits.

Collectively, these results demonstrate that stability components occupy distinct positions within structural space and are governed by different structural hierarchies. Complexity is shaped by distributed fine-scale topology, emerges from coordinated multivariate structural gradients, and robustness exhibits comparatively weak and representation-sensitive structural determination. Rather than reflecting a single structural driver, stability components are organized along layered and facet-specific dimensions of network architecture.

[Figure 8 about here.]

4 Discussion

4.1 Network structure is multidimensional

For several decades, ecological theory has sought to understand how the structure of food webs influences ecological stability. Yet despite extensive empirical and theoretical work, consensus has remained elusive. Different studies frequently identify different structural predictors of stability, and results often appear contradictory. Connectance has been linked both positively and negatively to stability, omnivory has been argued to stabilise or destabilise communities, and measures of modularity, trophic organisation, or motif structure have each been proposed as key determinants of ecosystem dynamics.

Our results suggest that much of this inconsistency may arise from a conceptual simplification: network structure is rarely a single property. Instead, food-web architecture appears to be organised along several coherent dimensions of variation. Structural metrics commonly used in ecological studies do not behave as independent descriptors, nor do they represent interchangeable proxies for a single structural attribute. Rather, they cluster into statistically robust modules that capture distinct aspects of network organisation.

This modular structure implies that ecological networks occupy a multidimensional structural space, in which different metrics describe different axes of variation in how interactions are arranged. In this view, commonly used descriptors such as connectance, trophic level, motif frequencies, or network distance are not competing measurements of the same property but rather partial projections of a more complex architectural system. The structure–stability debate has therefore often compared results derived from fundamentally different structural dimensions.

Structural modules reveal dimensions of network architecture

The clustering analysis revealed consistent groupings of metrics describing different aspects of food-web organisation. Some modules correspond to large-scale architectural expansion, integrating species richness, interaction density, motif frequencies, and global network properties. Others capture more specific aspects

of trophic organisation, such as the degree of vertical integration among trophic levels or the geometry of energy-transfer pathways.

The emergence of these modules indicates that food-web structure is not best understood as a flat collection of descriptors but as a hierarchically organised architectural system. Certain structural properties scale together as networks expand, while others vary independently and represent alternative ways of organising trophic interactions.

Importantly, the existence of these modules suggests that the large number of metrics currently used in network ecology largely represent a smaller number of underlying structural domains. Many descriptors capture overlapping information, whereas others represent distinct aspects of network organisation that are rarely considered together. Recognising these domains provides a principled basis for reducing metric redundancy and for interpreting structural patterns across ecosystems.

4.2 Reinterpreting the structure–stability debate

e.g., May 1972 vs. McCann 2000 looks like a contradiction. However results suggest they were simply looking at different modules. May's work focused on Return (spectral properties), while McCann focused on Persistence/Resistance (pathway dampening). By identifying these as independent axes in the PCA, we show that a network can be structurally complex (low return stability) but modularly dampened (high resistance).

The multidimensional nature of network structure has important implications for how stability relationships are interpreted. Much of the historical debate surrounding ecological stability has implicitly treated "structure" as a single explanatory variable. Studies typically select one or a small number of metrics and evaluate their relationship with a particular stability measure. However, if structural descriptors capture different architectural dimensions, then such analyses effectively examine different slices of structural space.

Under this interpretation, apparently conflicting results across studies become less surprising. A study focusing on connectance may capture variation in trophic integration, while another emphasising motif frequencies may instead reflect changes in network expansion and pathway complexity. Each metric may relate to stability through different mechanisms, yet comparisons between them have often been framed as competing tests of the same hypothesis. Rather than asking which structural metric best predicts stability, it may therefore be more productive to ask which dimension of network architecture is relevant to a given stability mechanism. Stability itself is a multidimensional property, encompassing processes such as persistence, resistance to disturbance, and the reorganisation of interaction pathways following perturbation. Different structural domains may

influence these processes through distinct ecological mechanisms. Our analysis does not attempt to resolve which particular metrics control specific stability outcomes. Instead, it highlights the need to reframe the problem: understanding ecological stability requires considering how multiple structural dimensions interact with different dynamical processes.

4.3 Implications for the design of network analyses

Recognising the multidimensional structure of food webs has practical implications for ecological analysis. First, it suggests that the common practice of selecting individual network metrics in isolation may obscure important structural relationships. Because many metrics capture overlapping aspects of network architecture, analyses based on a single descriptor may inadvertently reflect broader structural domains. Second, dimensionality reduction approaches provide a useful framework for representing network structure more coherently. By identifying modules of related metrics or latent axes of structural variation, researchers can characterise the major architectural gradients that define network organisation. Such approaches preserve the diversity of structural features while avoiding redundancy among descriptors. Finally, this perspective encourages a shift from metric-driven analyses toward process-driven structural representation. Rather than selecting metrics based on convention, analyses can instead focus on structural domains most relevant to the ecological process under investigation. This approach may provide clearer mechanistic insight into how network architecture shapes ecosystem dynamics.

4.4 Toward a structural framework for ecological networks

The results presented here suggest that the architecture of ecological networks can be understood as a low-dimensional structural system emerging from the interactions among species and their trophic roles. Within this system, individual metrics represent different projections of a broader architectural space. Recognising this structure does not eliminate the complexity of ecological networks, but it provides a way to organise and interpret that complexity more coherently. More broadly, this perspective highlights the importance of treating network structure as a multivariate ecological property rather than a collection of independent descriptors. Just as biodiversity is now understood through multiple complementary dimensions, including richness, composition, and functional traits, the structure of ecological networks may be best described through a set of interacting architectural domains. Future work linking network structure to ecosystem dynamics may therefore benefit from explicitly considering this multidimensional framework. By aligning structural dimensions with ecological processes and dynamical mechanisms, it may be possible to move beyond debates over individual metrics and toward a more unified understanding of how interaction networks shape the

stability of ecological systems.

5 Conclusion

The long-standing structure–stability debate in ecology may not reflect disagreement about ecological mechanisms, but rather the fact that network structure itself is multidimensional, and different studies have examined different dimensions of the same architectural system.

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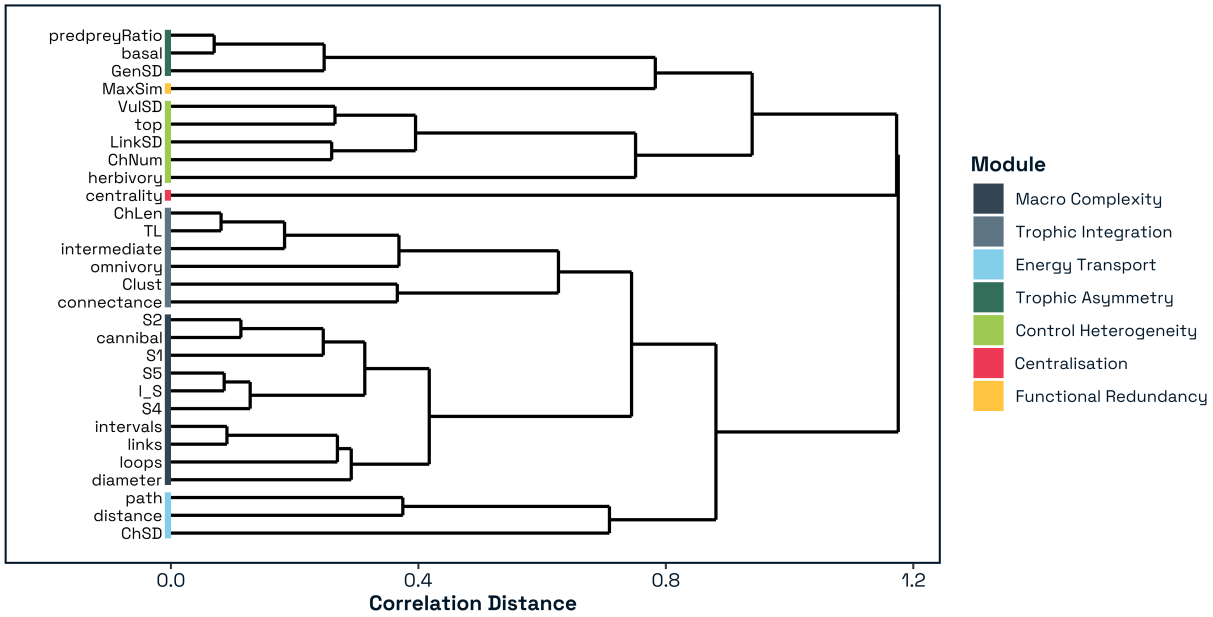

Figure 1: Hierarchical clustering of food-web structural descriptors. Dendrogram illustrating the modular organization of structural metrics based on a signed correlation matrix. Clustering was performed using UPGMA. Coloured bars indicate the seven distinct modules, reflecting robust association patterns among metrics. The horizontal axis represents the correlation distance (1−r), where shorter branches indicate stronger associations between descriptors.

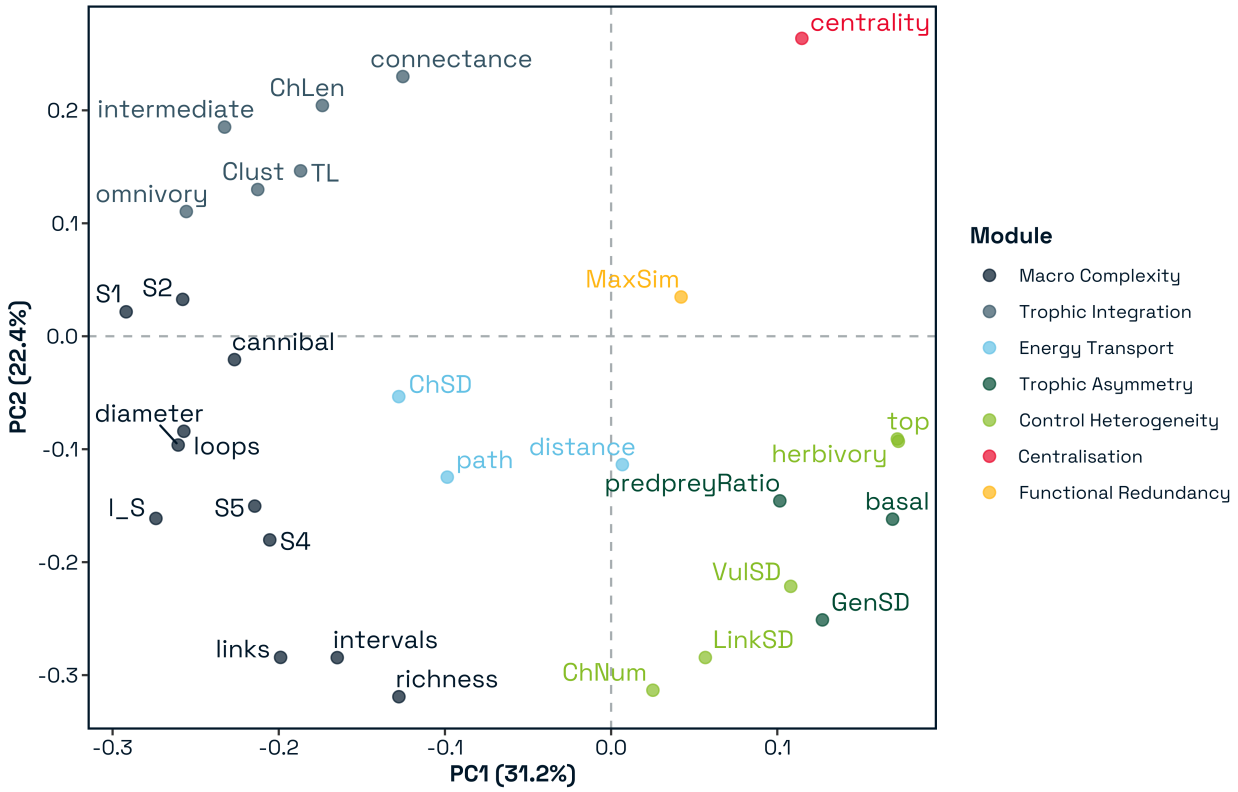

Figure 2: PCA loadings plot illustrating the dominant gradients of food web variation. Each point denotes a structural metric, with colours corresponding to the modular groupings. The clustering of like-coloured points along specific PC axes demonstrates that the modular organisation is preserved within a reduced dimensional space.

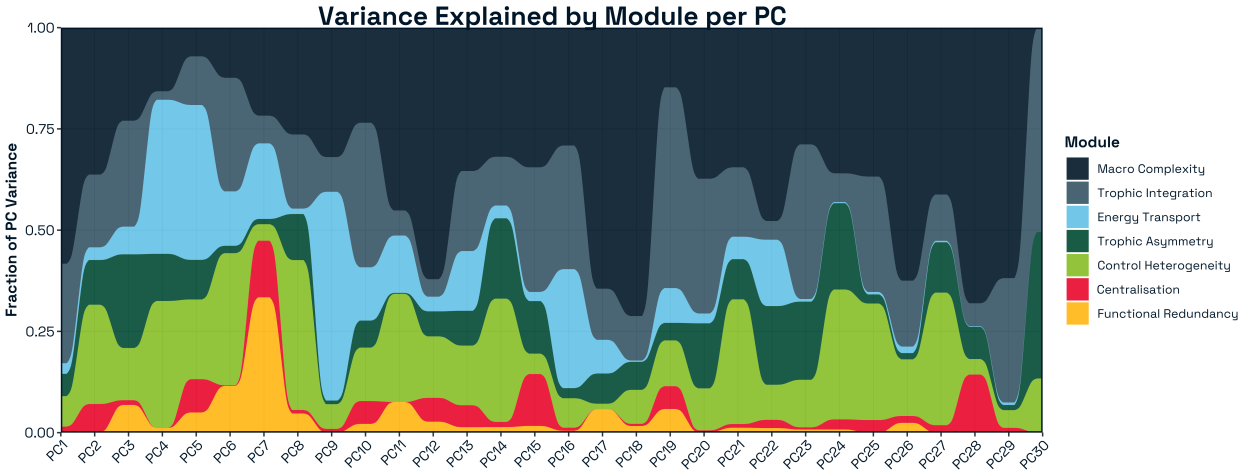

Figure 3: This figure quantifies the relative contribution of each structural module to the variance captured by individual Principal Components (PCs). Calculation: For each PC, the total variance (eigenvalue) is partitioned among the k=7 modules identified in the hierarchical clustering. The contribution of a module to a specific PC is calculated as the sum of the squared loadings of all metrics belonging to that module, normalized by the total variance explained by that PC. Visual Interpretation: The alluvial plot illustrates how the ecological weight shifts across different structural dimensions. For example, while PC1 may be dominated by a single module, subsequent PCs often represent a more diverse blend of modular contributions, reflecting the multifaceted nature of topological complexity and trophic organization. Significance: This visualisation confirms that the PC axes are not merely statistical artifacts but are grounded in the specific groups of correlated metrics defined in the main text. The stability of the flow across the first several components demonstrates that the modular organisation of food web architecture is consistently represented across the primary gradients of structural variation.

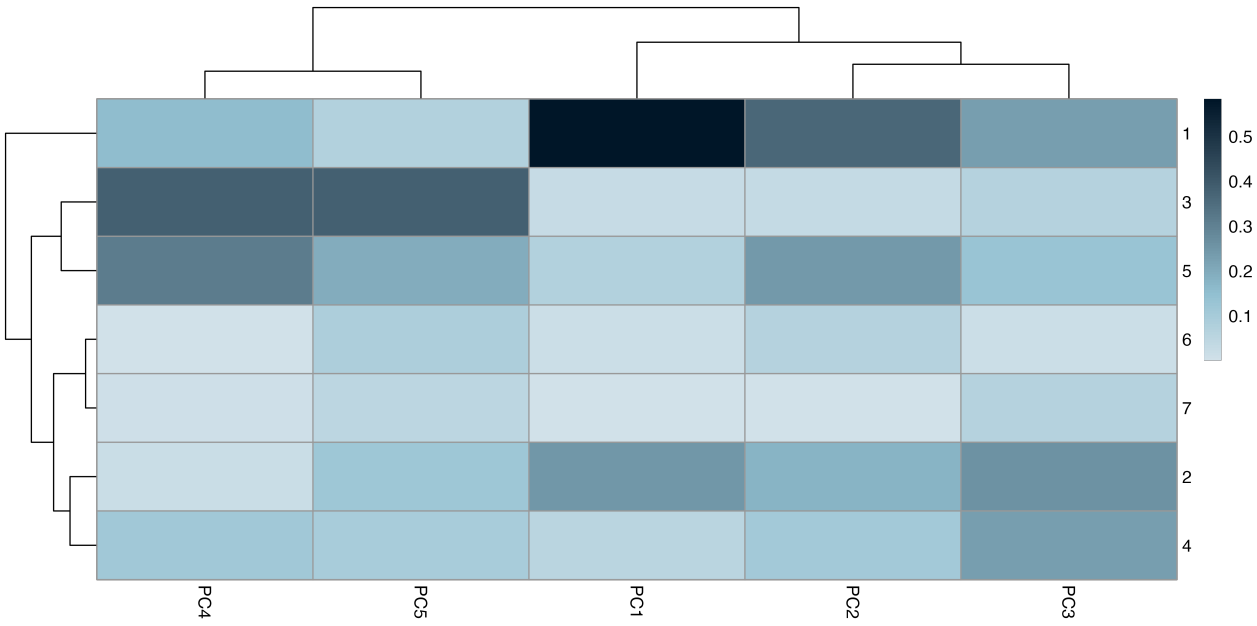

Figure 4: Heatmap showing the correspondence between the seven identified food-web modules (rows) and the principal components (columns) capturing 80% of cumulative variance. High values (darker shading) indicate strong alignment, where a specific module's internal structure drives the variation captured by that PC. Both rows and columns are hierarchically clustered to highlight similarities in how different structural dimensions contribute to the overall food-web architecture.

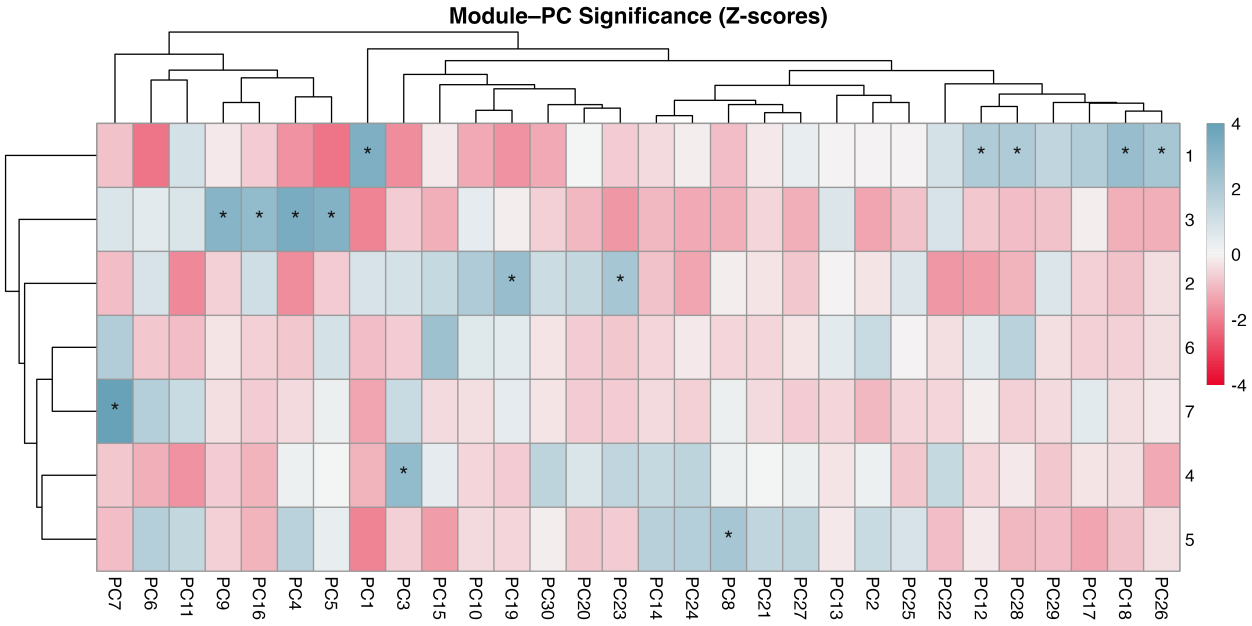

Figure 5: Heatmap showing module–principal component alignment quantified by permutation-derived Z-scores. Colour intensity reflects the magnitude and direction of alignment, and asterisks denote cells that meet joint significance criteria (permutation $P < 0.05$ and $|Z| > 1.96$). Rows correspond to structural metric modules and columns to principal components. Clustering highlights patterns of shared alignment across modules and multivariate structural gradients.

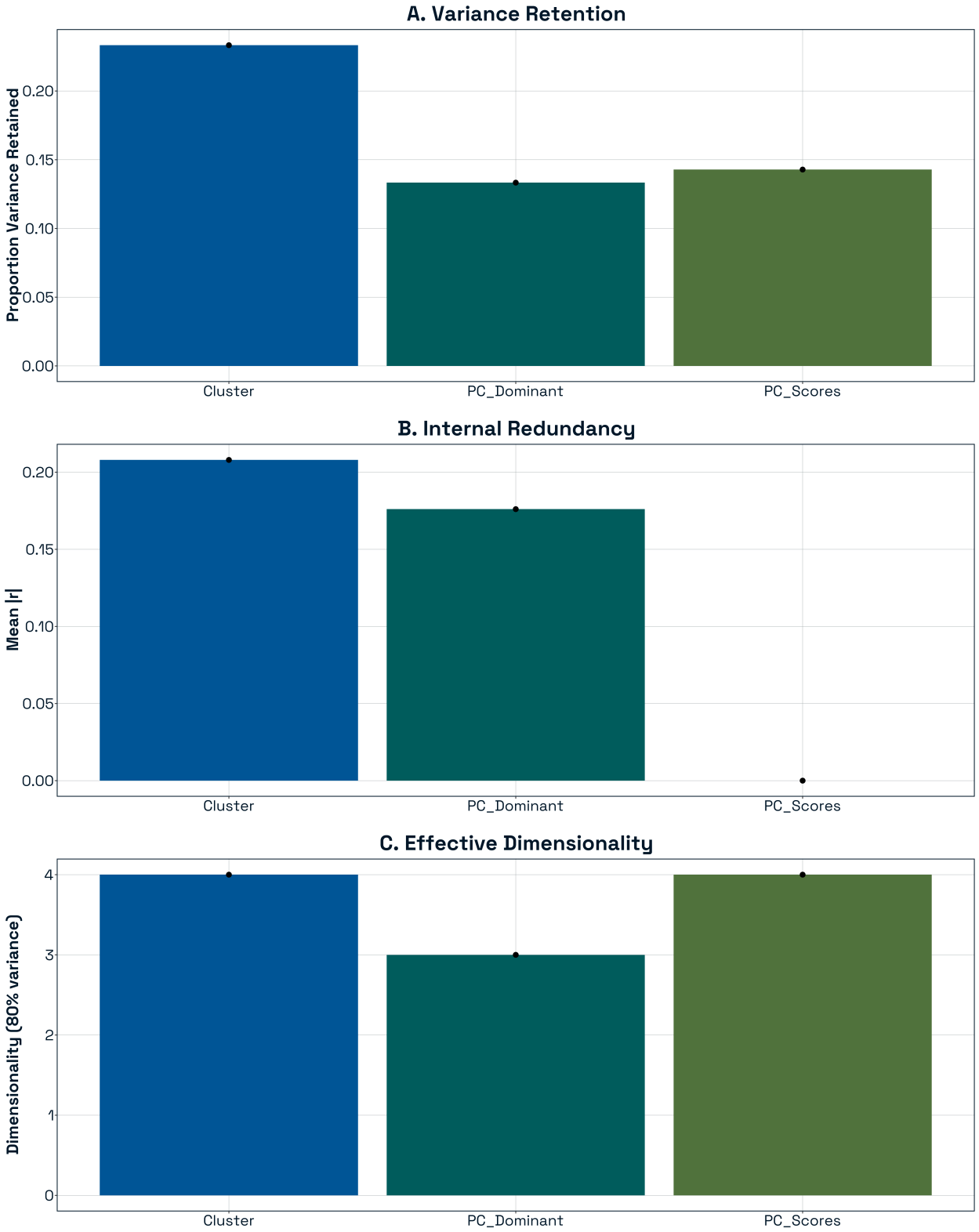

Figure 6: *Figure S1.* Characterisation of alternative dimensional-reduction approaches for network structural metrics. (A) Proportion of total variance in the full metric space retained by each representation: cluster medoids, PC-dominant metrics, and retained PC scores (80% cumulative variance). (B) Mean absolute pairwise correlation (|r|) among predictors within each representation, quantifying internal redundancy. (C) Effective dimensionality, defined as the number of axes required to explain 80% of variance within each reduced predictor set. Together, these diagnostics illustrate differences in information retention, redundancy, and structural alignment among dimensional-reduction strategies.

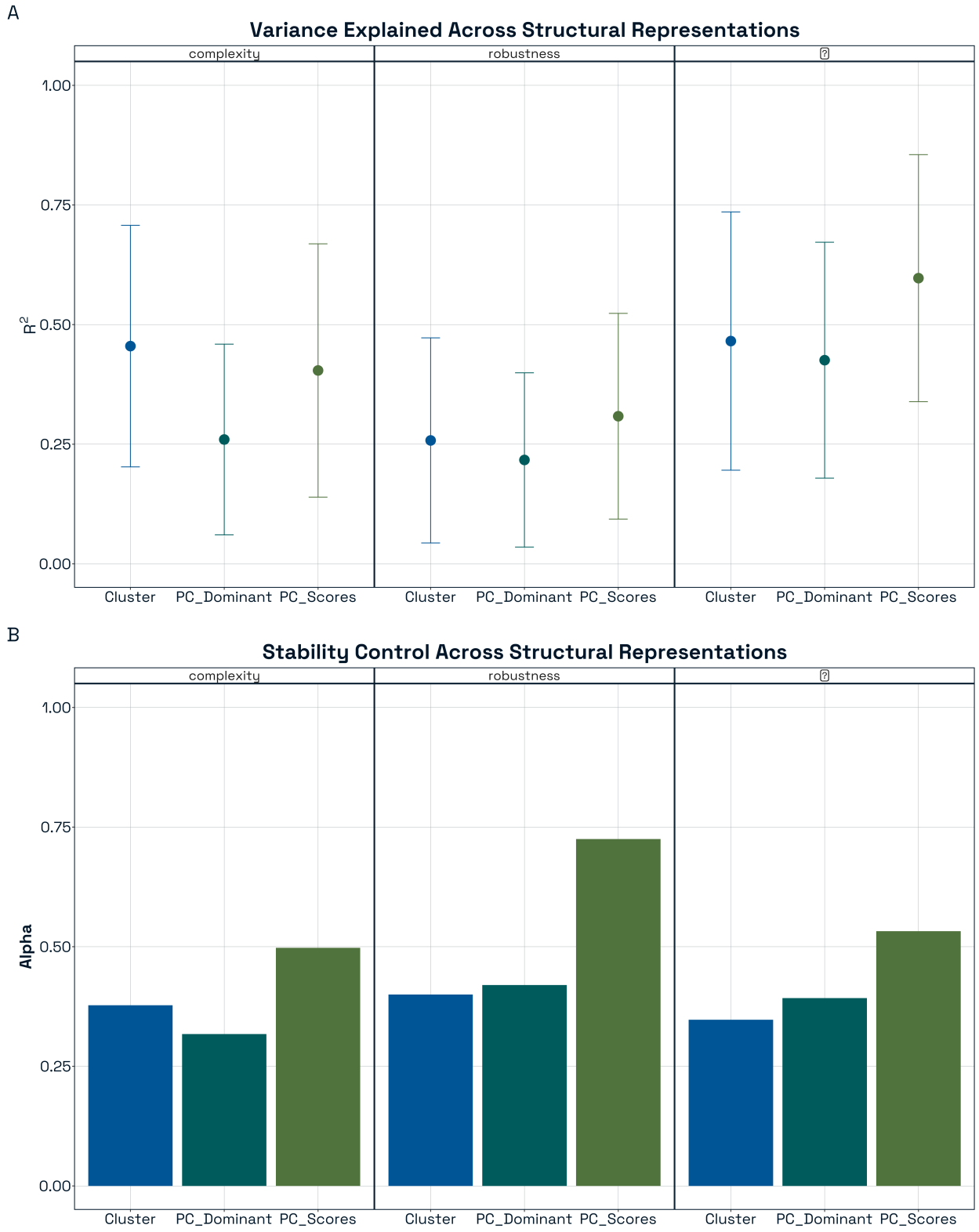

Figure 7: A: Predictive performance of structural representations across stability components. Cross-validated variance explained (R2) for each stability component (facets) under three representations of network structure: fine-scale cluster descriptors, dominant structural metrics, and principal component scores. Points represent mean predictive performance across repeated cross-validation folds; error bars denote ±1 standard deviation. Faceting separates stability components. Differences among representations indicate how the abstraction of structural space influences the strength of structural control across stability facets. B: Regularization structure across stability components and structural representations. Mean elastic-net mixing parameter ( ) across repeated cross-validation folds for each stability component and structural representation. Values near zero indicate ridge-like models where predictive signal is distributed across correlated predictors; values near one indicate lasso-like models where a small number of predictors dominate. Faceting highlights how structural

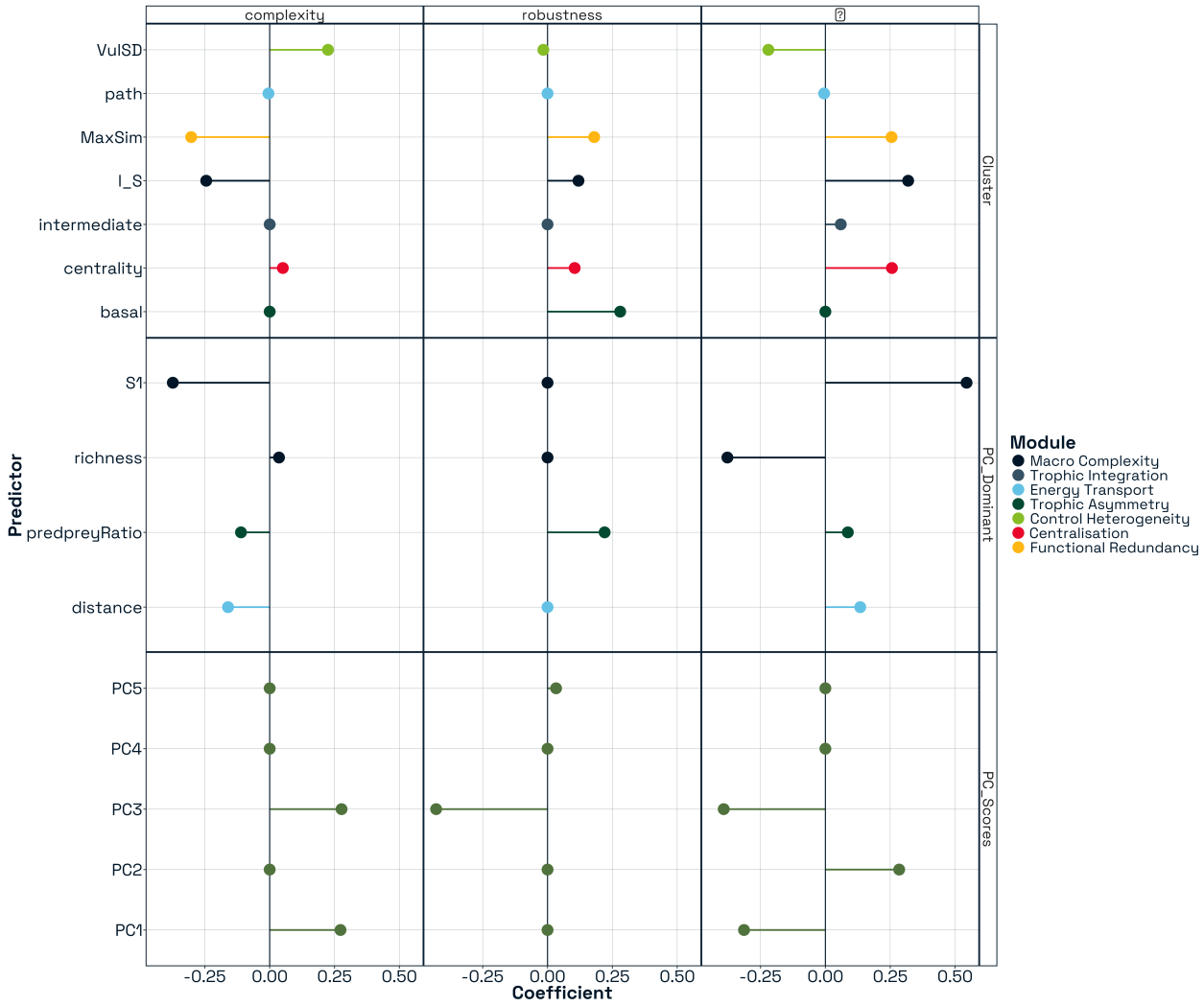

Figure 8: Standardised structural coefficients linking network predictors to stability components across representations. Elastic-net coefficients extracted from full models after cross-validation. Positive and negative coefficients indicate the direction and magnitude of structural influence. Predictor ordering reflects clustering metadata, and colours denote structural modules or metric groupings. Faceting by stability component and structural representation illustrates how the identity and strength of structural drivers reorganize as network space is represented at different levels of abstraction.